AberdeenGroup

Profile

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OmniBox^{SM} Interactive Broadcasting Network: Bringing Electronic Commerce And Entertainment To The Masses

Preface

Aberdeen Group believes that North American society is about to change in a fundamental way, as home consumers begin to tap the potential *of interactive television plus electronic commerce*. Intense communications-industry competition to 'own' the communications gateway into consumer's homes, demand for more — and more interactive — home electronic services, and new technology that can affordably deliver these services to the home in new ways, are driving the new market.

Numerous pilots by cable TV, telephone, direct-satellite and wireless-TV companies indicate that consumers want, and will pay for, more and better home services. Thus, Aberdeen foresees rapid delivery of the infrastructure needed to enable home consumers to buy or view what they want when they want, and to use their time more efficiently. Suppliers are closing in on the 'can-we-do-it' technical issues to the point of announcing delivery of first-generation home electronic-commerce offerings by mid-1997. But the devil will be in the infrastructure details: what's missing from many supplier announcements we have reviewed is a comprehensive architecture or model for delivering content, goods and services to the shopper of the future.

This *Profile* introduces OmniBox, Inc., a new industry player with a holistic vision of home electronic commerce and entertainment delivery, extensive patents behind their technology, and plans for rapidly delivering the key infrastructure components that are likely to accelerate the trend towards home electronic commerce via the Internet or interactive television.

Executive Summary

The question of 'Internet PC versus Interactive TV,' is becoming academic. It's a given that television will play a major role in home electronic commerce, because every household that cares to has one TV — if not more. At the same time, the communications infrastructure needs to augment, not supplant, the rapidly expanding home-computer market, especially in the area of home access to electronic mail and the Internet. Thus, whether by PC, TV or hybrid information appliances, Aberdeen believes that success in the nascent home electronic commerce market will ultimately hinge on the ability to engage and delight consumers. This will require more enticing Internet sites with greater audio/video content; more television channels to deliver a wider variety of content and services; a means to interact with the television's set-top box; and a secure mechanism for carrying out consumers' requests to order up a pay-for-view show, a new CD, or a silk camisole.

The *OmniBox*^{SM} *Network* is the most elegant and complete model we have seen so far for delivering new content and electronic commerce to home televisions and personal computers. Its key components are:

- *OmniBox-produced and -syndicated content* delivered via digital audio, pay-per-view, near-real-time video-on-demand and home shopping channels (e.g., interactive infommercials and videotext catalogs). OmniBox also has a patented way to deliver music albums electronically to the home — a key differentiator. With 28.8Kbps modems, most Internet applications are bandwidth-constrained and limited to static graphics. The *OmniBox Network* will be able to stream video content over ordinary phone lines into homes, creating new multimedia application opportunities.
- *A transaction processing network* enabling secure electronic ordering and payment via bi-directional cable or telephone to a financial services provider, which authorizes the transaction, places the order with the vendor, and credits the home-service provider with transaction commissions. OmniBox's intelligent TV remote, which works with

OmniBox-enabled set-top boxes and PCs, has a built-in, retail-store-quality, credit- and smart-card magnetic-stripe reader — a reliable, widely accepted means of authorizing pay-per-view or home shopping purchases;
- *Compression technology* shrinking video images up to 100-to-1 and enabling a whole new class of info-entertainment over the Internet, cable TV, or a satellite "head-end in the sky." For cable, OmniBox™ integrates analog and digital broadcasts, turning a single analog cable channel into numerous digital ones — increasing channel capacity a minimum of ten times. As multiple digital television services replace each precious analog channel, the system operator gains immediate access to a wealth of new content and advertising opportunities and to the transaction-based profit streams they generate.
- *An optional IBM-compatible-PC add-in board and software* endowing a PC with all of the functionality of a digital set-top box. PC owners can interact with *OmniBox Network* content using their monitors as TVs, or, with the appropriate software, over the Internet. Affluent households with PCs could thereby become an immediate market for satellite and cable providers who use OmniBox — before the market for digital set-tops matures in two years.

The *OmniBox*℠ *Network* Rollout

OmniBox plans to roll out its technology and services in mid-1997. The company will first target the Internet as a ready market for honing the delivery and services that comprise the *OmniBox Network*. At the same time, OmniBox is actively pursuing cable TV and other broadcast opportunities, huge markets that may take longer than the Internet to develop.

Table 1: OmniBox's Offering At A Glance

***OmniBox Network* Programming**
Satellite JukeBox℠ — Digital music sampling and downloading.
MailBox℠ — Electronic home shopping in real time.
VideoBox℠ — Digital video sampling-and-downloading, and pay-per-view events.
Services
Transactional and Order Processing — Patented process enables consumers to select and purchase virtually any product or service through their TV or PC.
Custom Turnkey Systems — Design and development of cost-effective solutions, including subscriber marketing programs, for converting existing capacity and services into full two-way interactive networks. Integrated system transmits third-party and OmniBox™ programming.
Equipment and Technology
Compression technology licensing/Head-end equipment — OmniBox's patented compression hardware and software work with industry standards such as MPEG-2 and can be licensed at very low cost by manufacturers and OEMs. Operators can control the digital encoding and broadcasting of their programming in real time at the local level.
Subscriber set-top boxes and remote control - High-performance/low cost set-tops based on patented compression technology include a reader for credit, debit or smart cards. and allow add-on devices such as CD-ROM players and recorders, printers and keyboards. A remote control option allows subscribers to select channels *and make purchases* from across the room.
PC add-in board and software - Provides the functionality of a digital set-top box so that PC owners can use their monitors as TVs and interact with OmniBox content.

Market Vision

OmniBox believes it can grab first an Internet ring, then the cable TV ring, and finally the brass ring of big-time success. While the *OmniBox Network* is ideally suited for cable operators, the cable companies have been slow to roll out digital TV offerings, constrained by the need to purchase millions of just-available set-top boxes. OmniBox management therefore plans first to deliver the *OmniBox Network* over the Internet to the 'Net's technology-savvy and largely upscale buyers. This first-phase offering will be accomplished in 1997 with OmniBox technology and:

- Close partnerships with Internet product suppliers such as PC manufacturers for large-volume distribution — the goal is to get OmniBox™ PC clients into over a million desktops and to build brand awareness;

- Content relationships that take advantage of the unique, video-intensive capabilities of the *OmniBox* ℠ *Network*. Content providers get a first-to-market chance to move the ‘Net beyond static graphics into a new-to-the-’Net text and video content format. Part of OmniBox’s value-add will be to establish a broadcast-quality Internet site and site builder;

- Technology partnerships with major broadcasting, switching, database and financial service providers to build and grow the *OmniBox Network* through its trial period.

Based on proprietary but architecturally open designs, the *OmniBox Network* satisfies the critical Internet-electronic commerce success factors of increased revenue opportunities, low time and cost to market, and technological flexibility:

- *Opportunities for system operators to participate in all revenue sources*. With the advent of home-based electronic commerce will come volumes of consumer transaction requests, requiring switching technology, billing systems and order processing experience that most ISP and cable operators lack. OmniBox’s patented transaction processing system allows the ISP or broadcast operator to receive a fee from these transactions without the headaches and cost of administering them at the head-end server.

- *Low time-and-cost-to-market*. The raison d’être of the *OmniBox Network* is to provide an infrastructure — and an audience — that advertisers and content providers will want to use and that can be rolled out rapidly to Internet, cable TV, telco, wireless and satellite subscribers. OmniBox expects to demonstrate its Internet technology in November. As the interactive TV market accelerates, the ramp-up for cable operators will be minimal because, with OmniBox’s analog-digital broadcast integration, they can add digital channels incrementally and gain immediate access to new profit streams without overhauling the existing analog plant. OmniBox was the first system supplier to successfully demonstrate this integration on Time Warner Cable’s Quantum system in New York City. OmniBox’s analog/digital capability reduces the operator’s set-top box and other basic component costs, yielding an attractive profit on each home in which the *OmniBox-Network* is installed.

- *Technological flexibility*. OmniBox focuses on deploying proprietary technologies as value added to a comprehensive, standards-compliant offering that emphasizes content and transaction processing capabilities. Thus, OmniBox™ content can be equally enjoyed with a light-weight Internet client or broadcast to its own cable hardware or to alternate set-top hardware from Primestar, Phillips and others. OmniBox’s technology is also compatible with the telephone lines, satellite, and microwave/wireless transmission vehicles, and could be available for delivery in those vehicles shortly after launch in the cable market.

Key Technologies

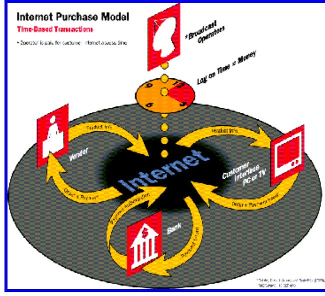
OmniBox holds fundamental patents, three of which figure prominently in the company’s value proposition: simple, secure transaction-processing; electronic software distribution; and low-cost compression.

OmniBox’s Transaction Processing System

Unlike current electronic commerce schemes that have not yet provided full security of credit card orders, OmniBox’s system uses the proven point-of-sale purchase model, where all transactions go from the consumer via a direct, private phone line to a financial institution for payment authorization (See Figure 1.) The system downloads images and text into the set-top (or streams catalog information to the PC) in a store-and-forward manner for browsing at the subscriber’s leisure. To buy a product, the subscriber selects from a menu and swipes a credit, debit or smart card through the reader located in the set-top box or smart remote control. Order and payment information are sent first to the financial institution, which verifies credit-worthiness, and then to the vendor along with payment authorization for shipment.

This approach should prove attractive to the financial institution because it provides the high levels of security and privacy of the U.S. -bank electronic funds transfer system. It also speeds payment clearance, minimizing "float", and offers a simpler, more robust two-network architecture. Research also indicates that subscribers will react favorably to the simplicity and security of this ordering process, resulting in increased pay-per-view transactions.

Figure 1: How OmniBox Supports Home-Based Electronic Commerce



Undisplayed Graphic

Electronic Software Distribution

The *OmniBox* ^{SM} *Network's* compression and store-and-forward capabilities create an opportunity for the company to pioneer the mass-market distribution of computer software, audio CDs, and even short videos. The encryption performed within OmniBox's compression algorithms solves a major problem of broadcast operators especially — the incidence of stolen signals for pay-per-view programming, which in some cases can be as high as *60 percent* according to Aberdeen research. Although untested, this market has potential in the $100 million-and-up market range by the year 2000. The two key enablers will be $300 (retail) CD-ROM recorders (estimated in 1998) and greater digital bandwidth into the home (1996 and after).

Low-Cost Compression

OmniBox's patented, digital video-compression technology reduces bandwidth requirements by a minimum factor of four versus MPEG and JPEG compression, while offering the same picture quality — or better — as on analog TV. An encoder and decoder incorporating this technology can be mass-produced by OmniBox, a manufacturer, or an OEM at a significantly lower cost than MPEG/JPEG. Cable operators can thus expand channels offered to subscribers and provide two-way interactivity more cost-effectively than their competitors without sacrificing picture quality.

Where OmniBox Is Especially Effective

Internet Service Providers (ISPs) — Fast, inexpensive Internet connections are an obvious use of OmniBox^{TM} technology. In the OmniBox architecture, users send an inquiry or request to the Internet via dial-up or cable modem. Typically, a 28.8Kbps modem is adequate for user-side data transfer. The benefit comes in the data download from the ISP. Information can be delivered via a telco, cable modem or even digital broadcast satellite. The unique *OmniBox Network* content, aided by video compression, enables a new class of customer-enticing Internet applications that could well blow the socks off of today's lush but static graphics environment. Of equal importance for companies evaluating a large Internet electronic commerce investment, the *OmniBox Network* offers an opportunity to trial a state-of-the-art content-delivery and secure transaction processing facility.

Cable TV Operators — OmniBox's offering has immediate benefits for cable operators with older coax plants and analog set tops that can provide only a limited number of higher-revenue-producing pay per view and premium channels. The most obvious applications for the additional digital channels obtained with OmniBox are extensions of existing home markets — video on demand, interactive retailing and information retrieval (downloads), video gaming, financial services and similar applications for which the TV rather than the PC will likely be the information appliance of choice.

Other Digital Broadcast Operators — OmniBox is also appropriate as a 'head-end-in-the-sky,' enabling the injection of OmniBox or local content for satellite, wireless and telco operators. The revenue stream would be split between OmniBox and the operator.

Corporate Applications — Large enterprises and IS organizations can use the *OmniBox*^{SM} *Network* as a software and multimedia-content distribution mechanism. Software manufacturers, for example, can mass-distribute product upgrades to thousands of customers in minutes without disks or shrink-wrapping — or electronically deliver to the retail point-of-purchase without having to stock actual copies of the products. Corporations can broadcast software, computer-based training, real-time video news feeds, or electronic files and documents to an unlimited number of locations at speeds (for large files) over 100 times that of today's typical transmission mechanism. Transmission fees are low enough to accommodate individuals and businesses of all sizes more effectively than point-to-point service. OmniBox can also support more cost-effective hotel video-on-demand, sports-bar direct-satellite feeds, or home movie-theater previews.

Competition — Real and Perceived

Although OmniBox faces competition in many of the product areas it plans to address (clearly, set-top boxes and cable modems have attracted big name competitors), we are aware of no other business venture with the broad scope, technology architecture, near-term deliverable products, and fundamental patents that OmniBox has.

OmniBox is distinguished from its nearest competitor, the Hughes DirecPC satellite information service (and Hughes' companion TV product, DirecTV) by its ability to span both TV and PC; its architectural flexibility across cable, phone line, and satellite; its electronic-commerce support; its breadth of value-added features; and its ability to support rapid third-party development of add-ons.

Providers of delivery systems — ISP, cable, direct broadcast satellite, and telephony — are themselves beginning to add the software and hardware to combine Internet and TV home-consumer applications. However, their present focus is either on implementing a particular high-speed communications technology — e.g., the cable operators with cable modems, the RBOCs with ADSL and ISDN — or on one or the other of the TV and PC markets. OmniBox differentiates itself here by its focus on higher levels of the communications infrastructure, network-delivery integration, its ability to support multiple high-speed media, its architectural flexibility, and its innovative compression and transaction-processing technologies. OmniBox is actually more a complement than a competitor to these suppliers' efforts.

Financials

OmniBox is a privately-held company that presently derives revenues from the direct sale of system hardware and software components, from royalties per transaction, from licensing the network to operators, and from residual royalties for preparing interactive programming for product vendors and advertisers. Its first round of funding was used to complete prototype development, secure patents, conduct field tests and establish working relationships. OmniBox is currently employing a second round of funding to support its national rollout scheduled for 1997.

Aberdeen Conclusions

In the home electronic-commerce world where Goliath-sized companies play, OmniBox has pulled off a tour de force — a broad spectrum of services in an elegantly simple architecture that rivals the most elaborate and expensive alternatives in its value proposition of:

- enabling interactivity to generate more revenues from new services and offering new services that leverage that interactivity;
- implementing elegant, low-cost technology that reduces the cost of digital conversion and the price point for set-top boxes; and
- leveraging satellite transmissions to eliminate problems of bandwidth, compatibility and line quality.

OmniBox's affordable solutions will enable even small operators to participate in the genesis of interactive TV. and home-based electronic commerce.

There will be lots of guns in this Wild West of markets, but we think the *OmniBox*^{SM} *Network* has a good shot not only at success but at delivering major new supplier and customer value-added across a wide range of vendor and consumer applications. Cable TV operators who are clamoring for more usable, revenue-generating channels will be particularly attracted to the OmniBox approach, which requires no rewiring from the head-end to the set-top box. But ISPs, direct satellite, telco, and wireless broadcast operators will also find attractive and unique offerings from OmniBox. Of the many computer and communications industry suppliers who are aiming at a profitable piece of a 100-million-household market in the United States alone, OmniBox stands out as an early leader.

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